

# Anurag Pandey

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## Education

UPES, Dehradun  
B.Tech C.S.E AI/ML

Aug 2023 – May 2027  
7.5 CGPA

## Professional Summary

Software Engineer and Machine Learning Engineer with hands-on experience building end-to-end applications — from training and deploying ensemble ML models to architecting REST APIs, databases, and browser extensions. Comfortable across Python, Java, and JavaScript with exposure to both full-stack development and production ML pipelines.

## Professional Experience

Notebrain Software & Services Pvt. Ltd  
AI/ML Intern

Aug 2025 – October 2025

- Built and evaluated ML models for sports outcome prediction integrated into a web application, owning the end-to-end pipeline from data preprocessing through feature engineering to model training and validation.
- Performed dataset cleaning and annotation across multiple data sources to prepare reliable training inputs, addressing missing values, inconsistencies, and class imbalance.
- Collaborated on iterative algorithm tuning and optimization in an agile workflow, contributing to measurable improvements in prediction performance.

## Venture Launcher

AI/ML Intern

June 2025 – July 2025

- Performed data analysis and preprocessing for an investor matching system, including data cleaning, feature identification, and exploratory analysis.
- Built a prototype ML model for matching relevance as a proof of concept for the founding team.

## Projects

### SurfVigil — Real-Time Browser Threat Detection Extension

- Built 4 threat-specific ML models — XSS detection (XGBoost, F1: **0.975**, AUC: **0.998**), phishing detection (Ensemble, F1: **0.995**, AUC: **0.999**), anomaly detection (IF + Autoencoder + RF hybrid, F1: **0.991**, AUC: **0.999**), and cryptomining detection (**94.7%** test accuracy) — each trained on browser-extractable features only.
- Architected FastAPI backend with parallel inference via ThreadPoolExecutor, achieving **~115ms** average end-to-end latency across all 4 models.
- Designed weighted ensemble scoring with hard-rule overrides and SHAP-based explainability for transparent Allow/Warn/Block decisions.
- Integrated full stack using object-oriented design: Chrome Extension (Manifest V3) → feature extraction (41 browser-level features) → backend ML inference → risk-based user actions.

### Smart City Guide — Java Swing Desktop Application

- Built a 4-layer Java desktop app (Presentation, Navigation, Services, Data Access) backed by a 7-table MySQL schema via JDBC, covering city profiles, tourism, transport, and user accounts across 20 cities.
- Implemented secure authentication using PBKDF2-HMAC-SHA256 (120K iterations, per-record salt, constant-time comparison); eliminated SQL injection surface via Prepared Statement on every query.
- Designed self-bootstrapping first-run flow (auto-creates schema, seeds data, persists credentials) and packaged v1.0 as a Windows installer with bundled JRE via jpackage — zero manual setup for end users.

### Rice Leaf Disease Detection System

- Built a two-stage ensemble of 5 models — Stage 1 (EfficientNet-B3, DenseNet-121, MobileNetV3) for 7-class triage at **96.68%** accuracy, Stage 2 (ViT-Base, ConvNeXt-Tiny) for bacterial disease refinement at **98.18%** accuracy.
- Optimized pipeline from 6 models to 5, reducing inference time by **16%** while improving accuracy by **0.91%**.
- Deployed end-to-end via PyTorch + Flask on Hugging Face Spaces with REST API, Docker support, and sub-2s processing time.

## Technical Skills

Languages: Python, Java, JavaScript, HTML, SQL, CSS

ML/DL: PyTorch, scikit-learn, XGBoost, Pandas, NumPy, ONNX Runtime

Backend & Databases: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB

Tools & Platforms: Git/GitHub, Docker, Hugging Face, Chrome Extension APIs (Manifest V3)

Core Concepts: Data Structures, Algorithms, OOP, Version Control, Model Deployment, Feature Engineering

## Position of Responsibility

### Core Member – UPES ACM & ACM-W Student Chapters

- Served as Program Head for *ICPC Code Nexus*, managing logistics and tech ops.
- Awarded Certificate of Appreciation for excellent execution and contribution.

### Social Outreach Intern — Shri Brahma Ji Siksha Avam Samaj Kalyan Samiti (May 2024 – July 2024)

- Coordinated education and environment awareness drives across 30+ communities, conducting 150+ field surveys.